

ior of humans and recent advances in the attention mechanism (Bahdanau et al., 2014) and the pointer networks (Vinyals et al., 2015), we propose a novel method to deal with the rare or unknown word problem. The basic idea is that we can see in many NLP problems as a task of predicting target text given context text, where some of the target words appear in the context as well. We observe that in this case we can make the model *learn to point* a word in the context and copy it to the target text, as well as *when to point*. For example, in machine translation, we can see the source sentence as the context, and the target sentence as what we need to predict. In Figure 1, we show an example depiction of how words can be copied from source to target in machine translation. Although the source and target languages are different, many of the words such as named entities are usually represented by the same characters in both languages, making it possible to copy. Similarly, in text summarization, it is natural to use some words in the original text in the summarized text as well.

Specifically, to predict a target word at each timestep, our model first determines the source of the word generation, that is, on whether to take one from a predefined shortlist or to copy one from the context. For the former, we apply the typical softmax operation, and for the latter, we use the attention mechanism to obtain the pointing softmax probability over the context words and pick the one of high probability. The model learns this decision so as to use the pointing only when the context includes a word that can be copied to the target. This way, our model can predict even the words which are not in the shortlist, as long as it appears in the context. Although some of the words still need to be labeled as *UNK*, i.e., if it is neither in the shortlist nor in the context, in experiments we show that this *learning when and where to point* improves the performance in machine translation and text summarization.

The rest of the paper is organized as follows. In the next section, we review the related works including pointer networks and previous approaches to the rare/unknown problem. In Section 3, we review the neural machine translation with attention mechanism which is the baseline in our experiments. Then, in Section 4, we propose our method dealing with the rare/unknown word problem, called the **Pointer Softmax (PS)**. The exper-

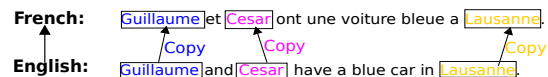


Figure 1: An example of how copying can happen for machine translation. Common words that appear both in source and the target can directly be copied from input to source. The rest of the unknown in the target can be copied from the input after being translated with a dictionary.

imental results are provided in the Section 5 and we conclude our work in Section 6.

2 Related Work

The attention-based pointing mechanism is introduced first in the pointer networks (Vinyals et al., 2015). In the pointer networks, the output space of the target sequence is constrained to be the observations in the input sequence (not the input space). Instead of having a fixed dimension softmax output layer, softmax outputs of varying dimension is dynamically computed for each input sequence in such a way to maximize the attention probability of the target input. However, its applicability is rather limited because, unlike our model, there is no option to choose whether to point or not; it always points. In this sense, we can see the pointer networks as a special case of our model where we always choose to point a context word.

Several approaches have been proposed towards solving the rare words/unknown words problem, which can be broadly divided into three categories. The first category of the approaches focuses on improving the computation speed of the softmax output so that it can maintain a very large vocabulary. Because this only increases the shortlist size, it helps to mitigate the unknown word problem, but still suffers from the rare word problem. The hierarchical softmax (Morin and Bengio, 2005), importance sampling (Bengio and Senécal, 2008; Jean et al., 2014), and the noise contrastive estimation (Gutmann and Hyvärinen, 2012; Mnih and Kavukcuoglu, 2013) methods are in the class.

The second category, where our proposed method also belongs to, uses information from the context. Notable works are (Luong et al., 2015) and (Hermann et al., 2015). In particular, applying to machine translation task, (Luong et al., 2015) learns to point some words in source sentence and copy it to the target sentence, similarly